Axles and alignment

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements and observations to improve straight rolling.

SUCCESS CRITERIA - I can...

1. Check axle parallelism and wheel clearance.
2. Record a repeatable alignment test.
3. Use measurements or observations to choose one adjustment.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “axle” and “alignment” to explain your answer. Sentence starters are optional.

axle: a shaft associated with a rotating wheel or gear

alignment: how the directions or positions of parts line up

clearance: space that lets neighboring parts avoid rubbing or colliding

01 Warm-up: bring back an earlier idea

A motor uses **5 W** for **7 s**. **Find electrical energy** and **name a possible non-mechanical output**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

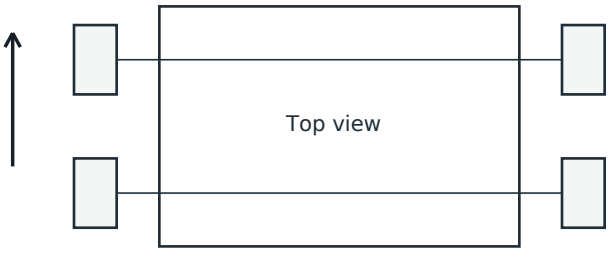
02 Worked example: explain the prediction

Reference: clearance = available gap - part width.
Use positive clearance and **check free rotation**. Alignment and wheel rubbing are separate checks; change one feature at a time.

Worked reasoning: Parallel axles can still cause drag if wheels rub the chassis. Alignment and clearance are separate checks; a straight top view does not verify free rotation.

Your turn: Draw a top view and a side detail marking an axle direction and a wheel gap.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

On your top-view sketch, mark both axle lines. **Add a check** for parallelism and **show clearance** between each wheel and the chassis.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

04 Apply the physics

A sample free-roll test ends **100 cm** forward and **8 cm** right of the starting line. Label that result. Plan three trials before and three after one alignment adjustment; **record actual measurements only after** testing.

Sentence starter: My design includes ... . It does what is needed by ... .

Design / trial	Forward (cm)	Side offset (cm)
Before 1	____	____
Before 2	____	____
Before 3	____	____
After 1	____	____
After 2	____	____
After 3	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A car veers right once. Does that prove the rear axle is crooked? **Name another possible cause** and a discriminating check.

Sentence starter: I notice ... . This supports ... because ... .

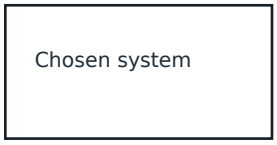
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 W** device runs for **6 s**. **Find transferred energy**. **Label input**, useful output and other energy transfers.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

Write the single adjustment you made and one observation that will tell you whether it helped.

Sentence starter: I notice ... . This supports ... because ... .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A car veers left. Propose two different causes and a test that separates them before changing both at once.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3